Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

Practical – 1

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Class: TYIT	Batch: 1
Date of Assignment: 24/07/26	Date/Time of Submission: 24/07/26

Self Study

```python
import pandas as pd
import matplotlib.pyplot as plt
```

[10] ✓ 0.0s

```python
df = pd.read_csv("NetFlix - NetFlix.csv")
print(df.head())
```

[14] ✓ 0.0s

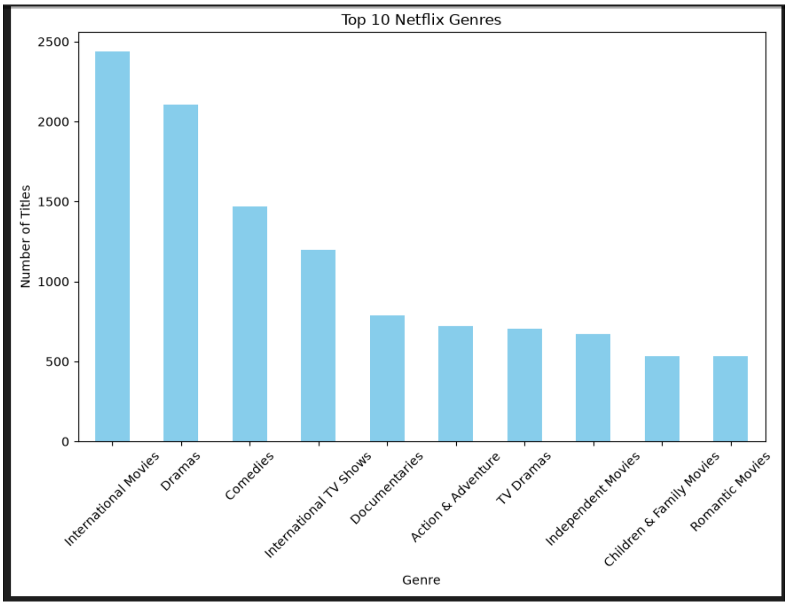
Q.1 Write a Python program to:

- Find the Top 10 Genres.
- Find the Bottom 10 Genres.
- Identify the most popular Genre.
- Identify the least popular Genre.
- Display the result using a bar chart.

Code:

```python
# Count genres
genre_count = df["genres"].str.split(", ").explode().value_counts()

print("Top 10 Genres")
print(genre_count.head(10))

print("\nBottom 10 Genres")
print(genre_count.tail(10))

print("\nMost Popular Genre")
print(genre_count.idxmax(), "-", genre_count.max())

print("\nLeast Popular Genre")
print(genre_count.idxmin(), "-", genre_count.min())

# Bar Chart
plt.figure(figsize=(10,6))
genre_count.head(10).plot(kind="bar", color="skyblue")
plt.title("Top 10 Netflix Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

Output:



Q.2 Write a Python program to find:

- The oldest Movie.
- The newest Movie.
- The oldest TV Show.
- The newest TV Show.

Display all details of these titles.

Code:

```
movies = df[df["type"] == "Movie"]
tvshows = df[df["type"] == "TV Show"]

# Oldest Movie
oldest_movie = movies.loc[movies["release_year"].idxmin()]

# Newest Movie
newest_movie = movies.loc[movies["release_year"].idxmax()]

# Oldest TV Show
oldest_tv = tvshows.loc[tvshows["release_year"].idxmin()]

# Newest TV Show
newest_tv = tvshows.loc[tvshows["release_year"].idxmax()]

print("Oldest Movie")
print(oldest_movie)

print("\nNewest Movie")
print(newest_movie)

print("\nOldest TV Show")
print(oldest_tv)

print("\nNewest TV Show")
print(newest_tv)
```

Output:

```
Output exceeds the size limit. Open the full output data in a text e
Oldest Movie
show_id                                                    s4961
type                                                       Movie
title                                             Prelude to War
director                                            Frank Capra
cast                                                         NaN
country                                            United States
date_added                                             31-Mar-17
release_year                                                1942
rating                                                     TV-14
duration                                                      52
genres                                Classic Movies, Documentaries
description      Frank Capra's documentary chronicles the rise ...
Name: 4402, dtype: object

Newest Movie
show_id                                                    s1286
type                                                       Movie
title                                                   Charming
director                                           Ross Venokur
cast             Wilmer Valderrama, Demi Lovato, Sia, Nia Varda...
country                       Canada, United States, Cayman Islands
date_added                                              8-Jan-21
release_year                                                2021
rating                                                     TV-Y7
...
duration                                                       4
genres                                     Kids' TV, TV Thrillers
description      A master thief who uses her skills for good, C...
Name: 250, dtype: object
```

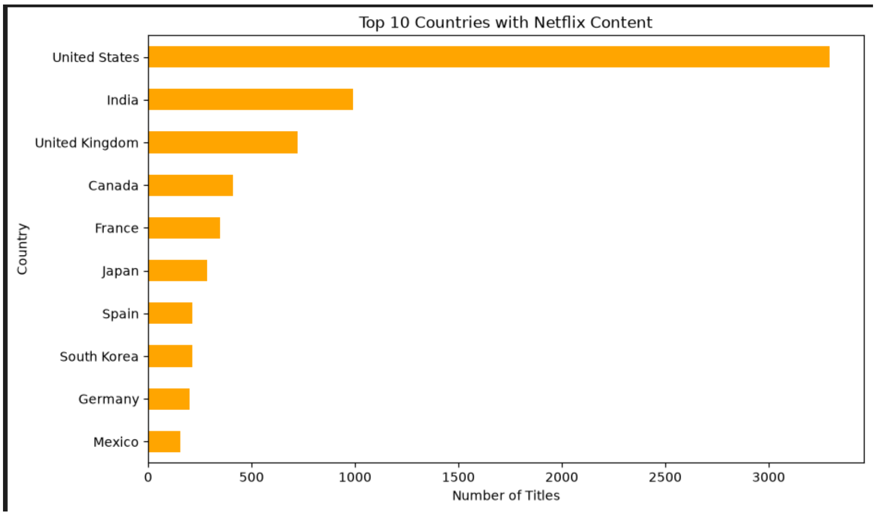
Q.3 Write a Python program to:

- Find the country with the highest Netflix content.
- Display the Top 10 Countries.
- Display the Bottom 10 Countries.
- Display the result using a horizontal bar chart.

Code:

```
country_count = df["country"].dropna().str.split(", ").explode().value_counts()

print("Country with Highest Netflix Content")
print(country_count.idxmax(), "-", country_count.max())

print("\nTop 10 Countries")
print(country_count.head(10))

print("\nBottom 10 Countries")
print(country_count.tail(10))

# Horizontal Bar Chart
plt.figure(figsize=(10,6))
country_count.head(10).sort_values().plot(kind="barh", color="orange")
plt.title("Top 10 Countries with Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.show()
```

Output:



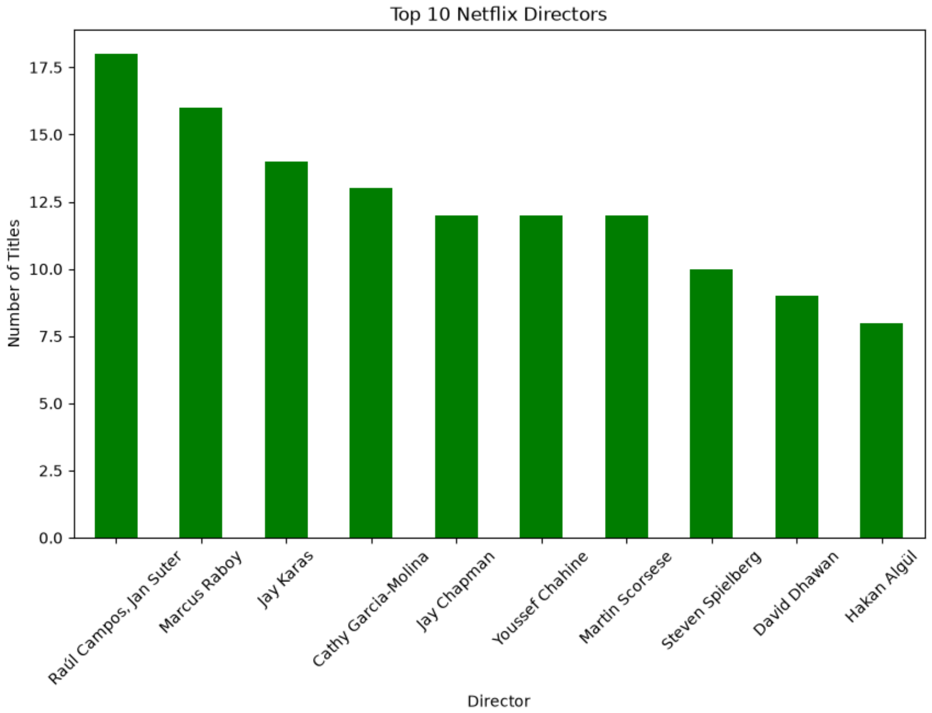
Q.4 Write a Python program to:

- Find the director with the highest number of Movies.
- Find the director with the highest number of TV Shows.
- Display the Top 10 Directors.
- Plot a bar chart of the Top 10 Directors.

Code:

```
director_df = df.dropna(subset=["director"])

movie_director = director_df[director_df["type"] == "Movie"]["director"].value_counts()

tv_director = director_df[director_df["type"] == "TV Show"]["director"].value_counts()

print("Director with Highest Number of Movies")
print(movie_director.head(1))

print("\nDirector with Highest Number of TV Shows")
print(tv_director.head(1))

print("\nTop 10 Directors")
print(director_df["director"].value_counts().head(10))

# Bar Chart
plt.figure(figsize=(10,6))
director_df["director"].value_counts().head(10).plot(kind="bar", color="green")
plt.title("Top 10 Netflix Directors")
plt.xlabel("Director")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```
✓ 0.1s

Output:



Q.5 Compare Netflix releases:

- Before 2020.
- During 2020.
- After 2020.
- Display the result using a bar chart or line chart.

Code:

```
before = len(df[df["release_year"] < 2020])
during = len(df[df["release_year"] == 2020])
after = len(df[df["release_year"] > 2020])

print("Before 2020 :", before)
print("During 2020 :", during)
print("After 2020 :", after)

release_data = {
    "Before 2020": before,
    "2020": during,
    "After 2020": after
}

plt.figure(figsize=(7,5))
plt.bar(release_data.keys(), release_data.values(), color=["blue","red","green"])
plt.title("Netflix Releases Comparison")
plt.xlabel("Release Period")
plt.ylabel("Number of Titles")
plt.show()
```

Output:

```
Before 2020 : 6888
During 2020 : 868
After 2020 : 31
```